

The sex chromosomes X and Y are involved in mammalian sex determination. Organisms with two wild-type (WT) X chromosomes typically develop as female (XX), but those with a combination of WT X and Y chromosomes typically develop as male (XY) (Table 1). Both sexes also inherit a set of autosomes (nonsex chromosomes). Humans inherit 22 pairs of autosomes (44 total). In contrast, rabbits inherit 21 pairs of autosomes (42 total).

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	X chromosome count	Y chromosome count	Autosome count
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Total chromosome population	3	1	84

Specifically, the *SRY* gene, normally located on the mammalian Y chromosome, is crucial for male sex determination and induction of testis development. Studies in mammals have indicated that, regardless of an organism's combination of sex chromosomes, fetuses always develop as female in the absence of the SRY protein. In addition, one of the X chromosomes becomes inactivated in WT,XX females to prevent them from having double the amount of X-chromosome gene products as WT,XY males.

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Researchers knocked out (KO) Sp1 binding sites on the *SRY* gene in rabbit embryos (Sp1 KO,XY rabbits) and observed their sexual development following implantation in surrogate mothers. The relationship between the *SRY* gene and sexual development can be explained by the following experimental observations:

- Most Sp1 KO,XY rabbits had female genitalia and displayed normal mating behaviors.
- Phenotypically female Sp1 KO,XY rabbits had significantly smaller ovaries and fewer follicles than WT,XX rabbits.
- No significant differences were observed in the cervix and uterus between Sp1 KO,XY and WT,XX rabbits.
- There was significantly reduced *SRY* expression in Sp1 KO,XY rabbits when compared with WT,XY males.

- WT,XY male rabbits had significantly higher testosterone levels when compared with Sp1 KO,XY rabbits.
- Sp1 KO,XY rabbits were infertile when mated with WT,XY males.
- Sp1 KO,XY rabbits could become pregnant via embryo transfer.

The number of each type of sex chromosome in a group of 5 male (WT,XY) and 5 female (WT,XX) rabbits is shown below.

	X chromosome count	Y chromosome count
5 Female	10	0
5 Male	5	5
Total sex chromosome population	15	5

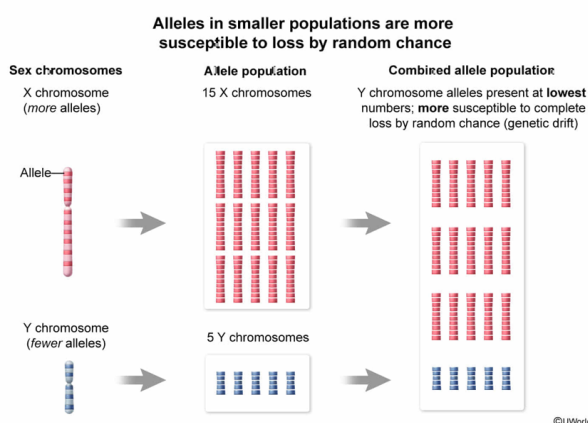
In the combined population of 15 X chromosomes and 5 Y chromosomes, which of the following types of allele, if any, would be most susceptible to complete loss from the population due to random chance alone?

- ✓ ☒ A. A Y-linked allele [56%]
- ☐ B. An X-linked allele [15%]
- ☐ C. Either an X- or a Y-linked allele [5%]
- ☐ D. Only an autosomal allele would be subject to complete loss by random chance alone. [21%]

Correct

57% answered correctly

Explanation:



Genetic drift is the fluctuation of **allele** frequencies within a population due to **chance** events. This can lead to random *loss* of alleles within a population. In general, compared to large populations, *small* populations are *highly* susceptible to genetic drift because small populations likely contain *lower* numbers of unique alleles in their **gene**

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pools. Consequently, genetic drift (ie, random allele loss) in small populations can eliminate alleles from the already small gene pool, decreasing the population's genetic diversity. Therefore, alleles in *smaller* populations are likely *more* susceptible to being lost due to genetic drift.

The sex of an organism is determined by the combination of **sex chromosomes** (XX or XY) that they inherit. Males receive an X from their mother and a Y from their father, whereas females receive two X chromosomes, one from each parent. Because the **Y chromosome** is present **only in males**, the Y chromosome population size is automatically *smaller* than that of the X chromosome.

In this scenario, the population size of each chromosome was analyzed in a group of 5 male (XY) and 5 female (XX) rabbits. The combined sex chromosome population is 20, with 15 X chromosomes (2 per female and 1 per male) and 5 Y chromosomes (1 per male).

Because the Y chromosome has the *smallest* effective population size, Y-linked alleles would be present at the *lowest* numbers in this population. Therefore, **Y-linked alleles** would be **most susceptible to complete loss** from the population due to **random chance alone** (ie, genetic drift).

(Choices B, C, and D) The probability of an allele being lost by random chance (ie, genetic drift) is influenced by population size. Because alleles exist on *all* chromosomes, alleles from *both* sex and nonsex chromosomes (ie, autosomes) *would* still be susceptible to loss by random chance alone. However, Y-linked (*not* X-linked) alleles would be most susceptible due to having the *smallest* population size.

Educational objective:

Genetic drift refers to random genetic changes in allele frequency that are due to chance events (not natural selection). Compared to large populations, small populations are *more* susceptible to allele loss via genetic drift.

Time spent:0 sec

Subject:
Biology

QID:401951

System:
Genetics and Evolution

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- No significant differences were observed in the cervix and uterus between Sp1 KO,XY and WT,XX rabbits.
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- WT,XY male rabbits had significantly higher testosterone levels when compared with Sp1 KO,XY rabbits.
- Sp1 KO,XY rabbits were infertile when mated with WT,XY males.
- Sp1 KO,XY rabbits could become pregnant via embryo transfer.

In an isolated population of 10,000 rabbits, 900 are homozygous for a recessive mutant *Sp1* allele. Assuming the population is in Hardy-Weinberg equilibrium, what is the percentage of rabbits in the population that would be heterozygous for the mutant *Sp1* allele?

- ☐ A. 9% [9%]
- ☐ B. 21% [23%]
- ✓ ☒ C. 42% [56%]
- ☐ D. 70% [10%]

Correct

56% answered correctly

Explanation:

Hardy-Weinberg Equilibrium

1. Determine alleles & possible genotypes

Alleles	Possible genotypes
S (wild-type dominant <i>Sp1</i>)	SS (homozygous dominant)
s (mutant recessive <i>Sp1</i>)	Ss (heterozygous)
	ss (homozygous recessive)

2. Use HWE equations to derive frequency of **Ss**

Allele frequency using HWE equation 1	Genotype frequency using HWE equation 2
$p + q = 1$	$p^2 + 2pq + q^2 = 1$
p = frequency of S q = frequency of s	p^2 = frequency of SS $2pq$ = frequency of Ss q^2 = frequency of ss

3. Use given information to calculate $2pq$

1. Solve for q using $q^2 = 900/10,000$
2. Use $p + q = 1$ to calculate p
3. Plug numbers into $2pq$ to get frequency of **Ss**

HWE = Hardy-Weinberg equilibrium.

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For populations in **Hardy-Weinberg equilibrium**, the allele frequencies can be used to predict the genotype frequencies using the following two equations:

- **$p + q = 1$** : Applies to a gene with only two alleles (S and s), where p represents the frequency of one allele (S) and q represents the frequency of the other (s).

- $p^2 + 2pq + q^2 = 1$: Can be used to calculate frequencies of possible genotypes (SS, Ss, and ss) from the allele frequencies. Here, p^2 and q^2 represent the frequencies of the two homozygous genotypes SS and ss, respectively. The frequency of the heterozygous genotype (Ss) is $2pq$.

In this case, heterozygous rabbits (Ss) have one dominant and one **recessive** mutant *Sp1* allele (ie, an autosomal allele), and the **frequency of the heterozygous genotype** in this population is equal to $2pq$.

To determine $2pq$, the following calculations must be performed:

1. Solve for q . Let p represent the frequency of the WT *Sp1* allele (S) and q represent the frequency of the recessive mutant allele (s). Here, 900 of 10,000 rabbits are homozygous recessive (ss) for the mutant allele. Given that q^2 is equal to the frequency of the homozygous recessive genotype (ss), the following solves for q :

$$q^2 = \frac{900}{10,000} = 0.09$$

$$\sqrt{q^2} = \sqrt{0.09}$$

$$q = 0.3$$

2. Use $p + q = 1$ to calculate p :

$$p + q = 1$$

$$p + 0.3 = 1$$

$$p = 1 - 0.3 = 0.7$$

3. Calculate the frequency of the heterozygous genotype (Ss) using $2pq$:

$$2pq = 2(0.7)(0.3) = 0.42$$

Therefore, the percentage of rabbits heterozygous for the mutant *Sp1* allele is **42%**.

(Choice A) 9% is the frequency of the homozygous recessive genotype (q^2).

(Choice B) 21% is half the frequency of heterozygotes in the population.

(Choice D) 70% is the frequency of the WT (dominant) *Sp1* allele (p).

Educational objective:

The Hardy-Weinberg equations can be used to relate allele frequencies to genotype frequencies. Allele frequencies are used to calculate the expected genotype frequencies of populations in Hardy-Weinberg equilibrium.

Time spent:0 sec

Subject:

Biology

QID:401952

System:

Genetics and Evolution

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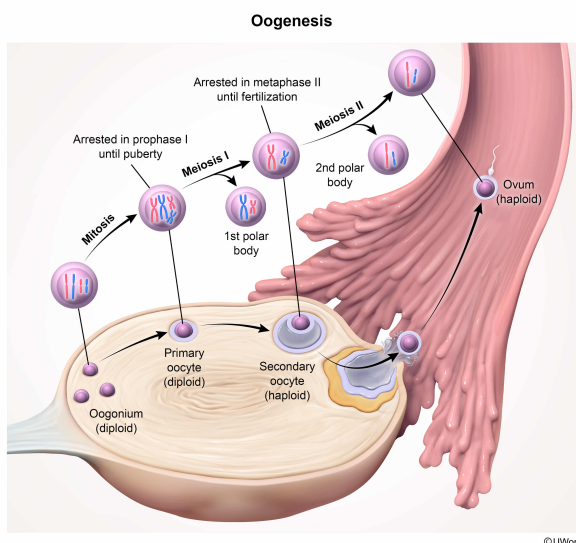
Based on the information given in the passage, which of the following observations best supports the conclusion that the second X chromosome that WT,XX females inherit affects their fertility?

- ☐ A. Sp1 KO,XY rabbits have lower testosterone levels compared with WT,XY males. [2%]
- ☐ B. Sp1 KO,XY rabbits and WT,XX rabbits exhibit no differences in uterus and cervix structure. [6%]
- ✓ ☒ C. Sp1 KO,XY rabbits have smaller ovaries and fewer follicles than WT,XX females. [88%]
- ☐ D. Sp1 KO,XY rabbits show reduced SRY protein expression compared to WT,XY males. [3%]

Correct

87% answered correctly

Explanation:



The **ovaries** are female reproductive structures that produce gametes and secrete sex hormones. Oogenesis or female gamete (ovum) production occurs in the ovaries as follows:

1. In utero, **oogonia** (ovarian stem cells) of the female embryo rapidly multiply via mitosis to generate **primary oocytes**, which are surrounded by specialized cells that form a saclike structure known as a *follicle*.

2. Female gametes must undergo meiosis to mature. Primary oocytes begin the first meiotic division but become arrested at prophase I until puberty.
3. At puberty, hormonal changes during each **menstrual cycle** result in a single follicle being selected to continue meiosis I. Completion of meiosis I produces one haploid **secondary oocyte** and one small polar body that ultimately degenerates.
4. The secondary oocyte begins the second meiotic division but is arrested at metaphase II.
5. In the ovulation phase of the menstrual cycle, the follicle ruptures and the secondary oocyte is released into the abdominal cavity.
6. The secondary oocyte enters the fallopian tubes, where it can be **fertilized** by a sperm cell. If fertilization occurs, the secondary oocyte will complete meiosis II to form one large **ovum** (fully mature) and a second polar body that degenerates.

Because the ovaries and ovarian follicles facilitate oogenesis, which is required for fertilization and pregnancy, **females** with **underdeveloped ovaries** or **reduced follicle count** may have **fertility issues**. The passage states that **Sp1 KO,XY rabbits** develop as female but have smaller ovaries and fewer follicles than WT,XX female rabbits, indicating that the second X chromosome may contribute to fertility by promoting normal ovarian and follicular development.

(Choices A and D) To determine whether the second X chromosome contributes to *female* fertility, the structure and function of reproductive organs in *females* with only one X chromosome (Sp1 KO,XY rabbits) must be compared to those in *females* with *two X chromosomes* (WT,XX rabbits), *not* to *male* (WT,XY) rabbits.

(Choice B) Based on the passage, Sp1 KO,XY rabbits and WT,XX females exhibit no differences in uterine or cervical structure. This finding supports the opposing conclusion that the second X chromosome has *no* effect on female fertility.

Educational objective:

The ovaries and ovarian follicles promote reproduction by facilitating the maturation and release of female gametes.

Time spent:0 sec
Subject:
Biology

QID:401953
System:
Genetics and Evolution

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- There was significantly reduced *SRY* expression in Sp1 KO,XY rabbits when compared with WT,XY males.
- WT,XY male rabbits had significantly higher testosterone levels when compared with Sp1 KO,XY rabbits.
- Sp1 KO,XY rabbits were infertile when mated with WT,XY males.
- Sp1 KO,XY rabbits could become pregnant via embryo transfer.

If researchers were to artificially insert the *SRY* gene in WT,XX female rabbit embryos using a vector, then these rabbits expressing *SRY* would most likely develop:

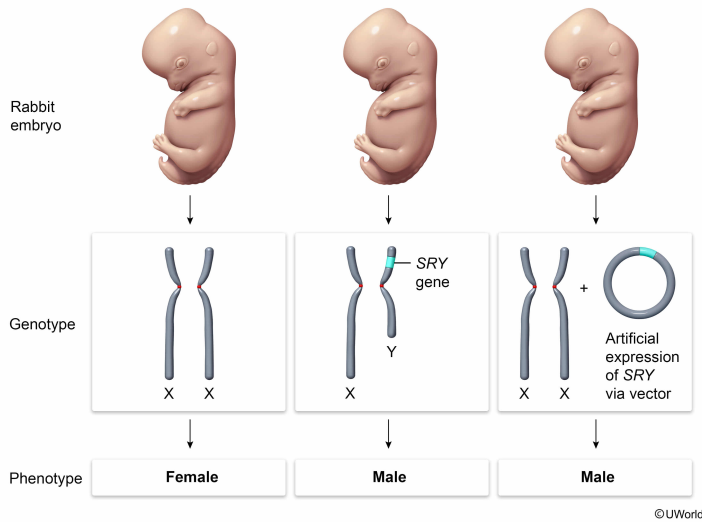
- ☐ A. male characteristics, because *Sp1* expression would increase and induce testis development. [9%]
- ✓ ☒ B. male characteristics, because expression of *SRY* would induce testis development. [72%]
- ☐ C. female characteristics, because the ratio of X to Y chromosomes dictates the sex in mammalian organisms. [4%]
- ☐ D. female characteristics, because X chromosome inactivation in WT,XX females is not affected by *SRY* expression. [13%]

Correct

72% answered correctly

Explanation:

SRY gene expression induces male development



According to the passage, the **SRY gene** is a mammalian **sex-determining gene**. *SRY* is normally found on the Y chromosome, which is present only in males, and its expression induces the development of the male gonads (testes). When the SRY protein is produced, fetuses develop as male, but in the absence of SRY, fetuses develop as female.

In this scenario, researchers artificially express the *SRY* gene in female rabbit embryos using a vector. Despite being genetically female (XX), the presence and expression of the SRY protein would likely **promote testis development**, causing these embryos to display **male sexual characteristics**.

(Choice A) According to the passage, Sp1 is a transcription factor that promotes *SRY* expression. However, the passage does *not* suggest that increased *SRY* expression increases *Sp1* expression. In addition, it is *SRY* (*not* Sp1) that is directly responsible for inducing testis development.

(Choice C) The passage states that sex in mammalian systems is determined by *SRY* expression, *not* by the ratio of X to Y chromosomes.

(Choice D) The artificial expression of *SRY* in genotypically female embryos likely induces the development of male, *not female*, characteristics. Although the passage includes discussion of X inactivation, the *SRY* gene is not normally present in female embryos; therefore, *SRY* expression does *not* affect X inactivation in WT, XX females.

Educational objective:

Mammalian sex is determined by the expression of the *SRY* gene on the Y chromosome. In males (XY), *SRY* expression induces the development of male sexual characteristics. In females (XX), the lack of *SRY* expression leads to the development of female sexual characteristics.

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Biology

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System:

Genetics and Evolution

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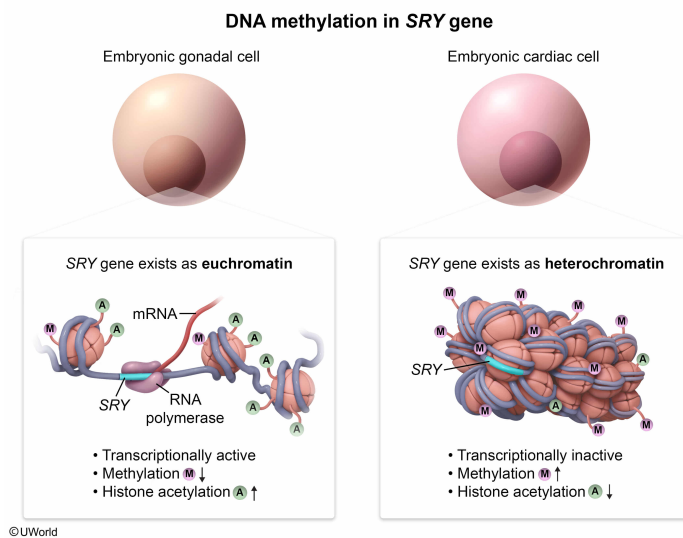
A gene expression assay shows that *SRY* gene expression is regulated in male embryos during development. Which of the following describes the DNA methylation pattern likely exhibited by the *SRY* gene in developing gonadal cells compared to developing cardiac cells in an embryo?

- ☐ A. The *SRY* gene has higher levels of methylation in embryonic gonadal cells. [20%]
- ✓ ☒ B. The *SRY* gene has lower levels of methylation in embryonic gonadal cells. [55%]
- ☐ C. The *SRY* gene has similar levels of methylation in both cell types. [2%]
- ☐ D. The comparison cannot be performed because cardiac cells do not carry the *SRY* gene. [20%]

Correct

56% answered correctly

Explanation:



All cells that comprise a multicellular organism contain **identical genomes**. These cells can differentiate (ie, specialize in structure and function) during development through differential regulation of gene expression.

Chromatin structure (heterochromatin or euchromatin) affects gene expression by regulating **transcription**. **Heterochromatin** is tightly wound DNA characterized by *low levels of histone acetylation* and *high levels of DNA methylation*. Genes in these regions are considered *silenced* because the DNA is so tightly packed that it is inaccessible to the transcription machinery. In contrast, **euchromatin** is marked by *high levels of histone acetylation* and *low levels of DNA methylation*. This makes euchromatin looser and the DNA more accessible to the transcription machinery, allowing gene expression.

In male embryos, gonadal cells develop into the testes (the site of **spermatogenesis**), whereas cardiac cells develop into heart tissue. As described in the passage, the *SRY* gene is crucial for male sex determination, meaning it is most likely highly expressed in cells important to male sexual development (gonadal cells). In contrast, cardiac cells do not have a role in male sexual development.

Consequently, the ***SRY* gene in gonadal cells** likely exhibits **higher expression** than the *SRY* gene in cardiac cells and exists in the form of **euchromatin** (ie, with **lower levels of DNA methylation** and higher levels of histone acetylation).

(Choice A) In the male embryo, gonadal cells *express* *SRY* but cardiac cells do not. Accordingly, the *SRY* gene region in gonadal cells must be accessible to the transcription machinery and therefore exhibits *lower* levels of DNA methylation compared with cardiac cells.

(Choice C) If the genomic region containing the *SRY* gene had similar levels of DNA methylation in both cell types, this would indicate that levels of *SRY* gene expression are similar in both cells. However, because gonadal cells (*not* cardiac cells) are involved in male sexual development, *SRY* gene methylation and expression would be expected to differ in the two cell types.

(Choice D) All cells that comprise a multicellular organism contain the same DNA (ie, genome). Therefore, the *SRY* gene is present in *all* cells, and this comparison *can* be made.

Educational objective:

Heterochromatin is tightly packed DNA that is inaccessible to transcription machinery, preventing gene expression. Euchromatin is loosely packed DNA that is accessible to transcription machinery, allowing gene expression.

Time spent:0 sec

Subject:
Biology

QID:401955

System:
Genetics and Evolution